look even more different than we could ever imagine. Technology in the form of the Auto Pilot is already flying planes all across the globe, how soon before it's driving your car? Based on research already being done, and Google's driverless car initiative is an example.

Assembly lines are already mostly automated, but how long before we see automated waiters, cleaners, bakers, and more? In fact, technology does the job better – take micro circuit boards and CPUs as an example: no human could ever manufacture those, because they require nanometer precision that we're just not equipped for. Easily available robotic surgery will be next, and who knows, perhaps the human race will finally exist only to think – until something passes the Turing Test, that is. Then what?

## Sensors

For ages we've worked towards going beyond what human senses can distinguish. Hearing more accurately, seeing more clearly, or more precisely, smelling not just odours, but identifying the components in the air. All of this, and much more is done by sensors. The simplest and most common sensor we all interface with is a thermometer (or a temperature sensor of some sort). Whether it's taking your temperature to check for a fever, or adjusting the temperature on your air conditioner, it's all sensors at play.

And that's not all. Everywhere you go, all around you, in every nook and corner a sensor is working away furiously to make your life easier.

If you've ever been to a mall and have used those automatic doors, they're sensor driven. Cars have sensors for everything. Almost all warning signals – door open, seat belt, indicators, overheating, low fuel level, etc. – are sensor driven. Even your PC and phone are filled to the brim with sensors, and some of these are the first type we want to look at.

### Accelerometers and gyroscopes

Every time you turn your smart phone around, and you find the screen rotating to ensure you don't have to read upside down, you have just been sensed! Gyroscopes sense the orientation of a device, so it always knows where up is, and which direction is down. Thus you have programmers that set tolerances, but code devices to switch screen orientation in a certain way when the gyroscope reads changes by some set amount. Thus, you hold your phone upside down, and the screen does a 180 too.

Accelerometers are a little different, because they measure linear movement, usually between two axes. Even when your phone's on a table, not being moved, the accelerometer will recognise gravity as a force acting on it, but show zero linear movement. The Gyroscope will know whether the phone's face down or being balanced on its side, etc., and will decide whether the screen should orient this way or that. However, if you slide your phone along the table, the gyroscope may notice no difference, but the accelerometer will know that the phone accelerated for x seconds by y metres per second along